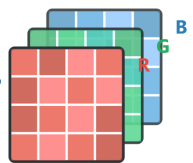
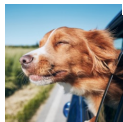


$\mathbf{x}_{\text{raw}} \in \mathbb{R}^{32 \times 32 \times 3} \rightarrow \mathbf{x}_{\text{flat}} \in \mathbb{R}^{3072}$

Flatten()



Normalize()

$$b_{d,k} = \begin{cases} 1 & \text{if } \hat{x}_d > t_k \\ 0 & \text{otherwise} \end{cases} \quad b_{d,k} \in \{0, 1\}$$

$$\mathbf{b} \in \{0, 1\}^{D \cdot K} \quad (D=3072, K=\text{resolution})$$

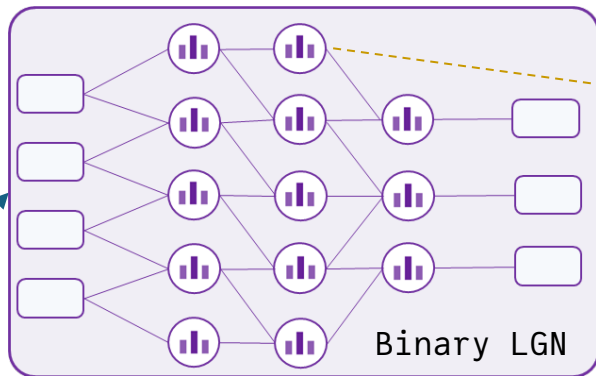
$$\boldsymbol{\tau} \in \{-1, 0, 1\}^{D \cdot K}$$

$$\tau_{d,k} = \begin{cases} +1 & \text{if } \hat{x}_d > t_k + \delta \\ 0 & \text{if } t_k - \delta \leq \hat{x}_d \leq t_k + \delta \\ -1 & \text{if } \hat{x}_d < t_k - \delta \end{cases}$$

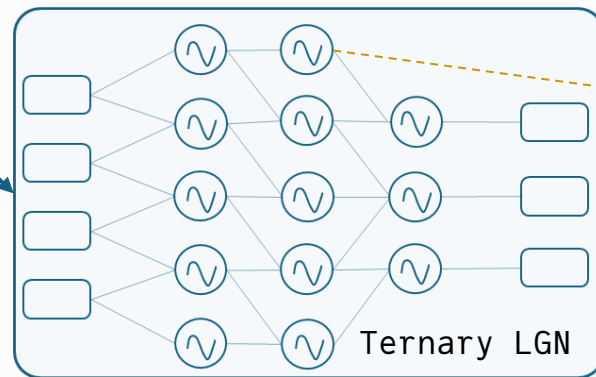
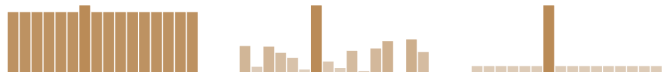
Polynomial
Surrogate
Training
(Ours) →

$$\delta = \frac{1}{2} \min_k (t_{k+1} - t_k)$$

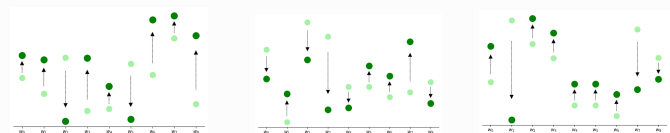
Softmax-over-gates



Binary LGN



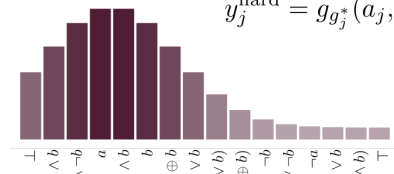
Ternary LGN



$$p(a, b) = w_0 + w_1 a + w_2 b + w_3 ab + w_4 a^2 + w_5 b^2 + w_6 a^2 b + w_7 a b^2 + w_8 a^2 b^2$$

$$g_j^* = \arg \max_{g \in \{0, \dots, 15\}} \ell_{j,g}$$

$$y_j^{\text{hard}} = g_{g_j^*}(a_j, b_j)$$



$$y_j^{\text{soft}} = \sum_{g=0}^{15} g_g(a_j, b_j) \cdot \sigma(\ell_{j,g})$$

$$\sigma(\ell_{j,g}) = \frac{e^{\ell_{j,g}}}{\sum_{g'=0}^{15} e^{\ell_{j,g'}}}$$

$$\mathcal{L}_{\text{binary}} = \text{CE}(\text{GroupSum}(\mathbf{y}^{\text{soft}}), \mathbf{y}_{\text{true}})$$

0.02	-0.9	0.05
-1	0.01	0.32
0.98	0.94	-0.9

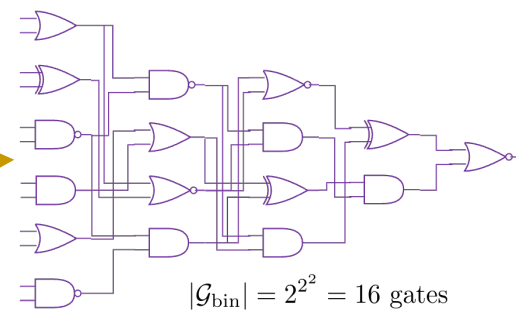
$$\mathbf{w}_j^{\text{hard}} = V^{-1} \text{round}_3(V \mathbf{w}_j)$$

0	-1	0
-1	0	1
1	1	-1

$$y_j^{\text{soft}} = \text{clip}(\mathbf{w}_j^\top \mathbf{m}(a_j, b_j), -1, 1)$$

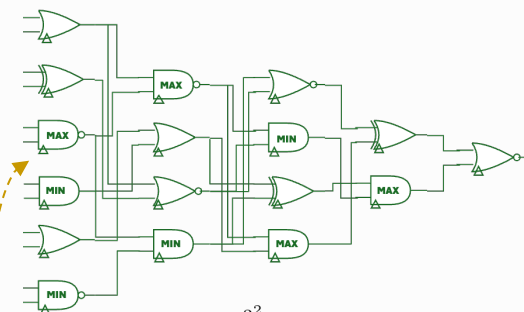
$$\mathbf{m}(a, b) = [1, a, b, ab, a^2, b^2, a^2 b, ab^2, a^2 b^2]^\top$$

$$\mathcal{L}_{\text{PST}} = \mathcal{L}_{\text{task}} + \lambda(s) \cdot \mathcal{L}_{\text{commit}} \quad ; \quad \lambda(s) = \lambda_{\text{max}} \cdot \left(\frac{s}{S}\right)^\gamma$$



Binary Logic IC

$$|\mathcal{G}_{\text{bin}}| = 2^{2^2} = 16 \text{ gates}$$



Ternary Logic IC

$$|\mathcal{G}_{\text{tern}}| = 3^{3^2} = 19,683 \text{ gates}$$

Input Encoding

Training Network

Hardening

Inference Circuit